High-Stakes Activation Probes on Indonesian

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Abstract

McKenzie *et al.* [10] trained linear probes on language model activations to detect high-stakes interactions. We test whether these probes generalize to a language absent from training by translating the evaluation datasets into Indonesian and evaluating without retraining. On Llama 3.1 8B (layer 12), mean AUROC on out-of-distribution benchmarks drops from 0.9046 to 0.8957 (0.9%, 10-seed mean). On Gemma 3 12B (layer 32), the gap is 0.8972 to 0.8873 (1.0%). Per-example agreement is high on Llama (91.7% stable predictions on Anthropic HH) and lower on Gemma (88.8%), indicating that aggregate AUROC can mask per-example inconsistency. Using Sparse Autoencoders to decompose probe failures on Llama, we find that error-enriched features correspond to broad domain categories (technology, finance) but do not identify specific failure mechanisms. On Gemma, the SAEs we tested did not produce diagnostic features. The cross-lingual degradation is small relative to the out-of-distribution gap between synthetic training data and naturalistic benchmarks (6–13%), which remains the more pressing concern.

1 Introduction

McKenzie *et al.* [10] introduced activation probes for detecting high-stakes LLM interactions, situations involving medical advice, legal counsel, financial decisions, or safety-critical scenarios. Their linear and attention probes, trained on residual stream activations, achieve strong performance on synthetic evaluation data but show degradation on naturalistic out-of-distribution benchmarks (Anthropic HH, ToolACE).

One practical question for deployment: do these probes work when users speak languages not represented in training data? If the probe relies on surface-level linguistic features, it would fail on new languages. If it captures a language-invariant concept in the model's representation space, it should transfer.

We test with Indonesian (Bahasa Indonesia), a language with approximately 200 million speakers that is typologically distant from the English/French/German/Hindi mix in the training data. Llama 3.1 8B [6] lists Indonesian as a Tier 2 supported language; Gemma 3 12B [4] includes Indonesian in its broader multilingual training. We evaluate on both models to check whether findings hold across architectures.

What we did. We translated all three evaluation sets to Indonesian, extracted activations from both models, and evaluated the trained probes without retraining. We swept layers on both models and compared early versus late layers. We then used Sparse Autoencoders (Llama Scope [7] and Gemma Scope 2 [5]) to investigate whether decomposed features could explain why probes fail on specific examples.

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What we found.

- Cross-lingual degradation is small: 0.9% mean AUROC on Llama, 1.0% on Gemma.
- Per-example agreement is high on Llama (91.7% stable) and lower on Gemma at layer 24 (88.8%), showing that aggregate scores can mask per-example inconsistency.
- The synthetic-to-real domain gap (6–13%) is the dominant concern, far larger than the cross-lingual gap.
- SAE decomposition on Llama identifies domain-level correlates of error (technology, finance). On Gemma, the SAEs we tested did not produce diagnostic features.

2 Related Work

Activation probes for safety-relevant properties. McKenzie *et al.* [10] train linear and attention probes on LLM activations to detect high-stakes interactions, achieving mean AUROC of 0.91 on Llama-3.3-70B.

Multilingual representations in LLMs. Wendler *et al.* [12] show that multilingual transformers converge to a shared representation space in middle layers, with language-specific encoding concentrated in early and late layers.

SAE-based interpretability. Sparse Autoencoders decompose model activations into interpretable features [2]. Llama Scope [7] provides SAEs for Llama-3.1-8B across all layers; Gemma Scope 2 [5] extends this to Gemma 3 models. Jiang *et al.* [8] propose SAE-derived embeddings as a general data-analysis tool. We use SAEs not to interpret model behaviour in general, but to decompose the specific failure modes of a downstream probe.

3 Setup

3.1 Methodology

We follow McKenzie *et al.* [10]:

- **Activations.** Residual stream at `input_layernorm` (before layer norm), mean-pooled over sequence length, with leading BOS token stripped. Max sequence length: 8,192 tokens.
- **Probe.** `StandardScaler` + `LogisticRegression` (L2, $C{=}10^{-3}$, `fit_intercept=False`, LBFGS solver, `max_iter=1000`).
- **Training data.** 8,000 examples (4,000 high-stakes, 4,000 low-stakes) from the paper's `prompts_4x` training set ($\approx$61% EN, $\approx$14% HI, $\approx$13% DE, $\approx$13% FR).
- **Seeds.** 10 random seeds (master seed 20250214); results are means with 95% bootstrap CIs. The L2-regularised LBFGS solver converges to the same optimum regardless of seed ($\sigma{=}0.000$ across all 10 seeds), so multi-seed serves as a sanity check rather than a variance estimate.
- **Models.** Llama 3.1 8B and Gemma 3 12B, loaded in 8-bit quantisation via bitsandbytes [3]. The paper uses fp16; we use 8-bit to fit on a single A10 GPU (24 GB).
- **Layer sweep.** Llama layers [12, 16, 20, 26, 31] of 32; Gemma layers [8, 16, 24, 31, 32, 41, 47] of 48.

We evaluate the linear probe only. The paper's best results use an attention probe on Llama-3.3-70B, which we did not run due to compute constraints.

3.2 Datasets

McKenzie *et al.* [10] generated the training and synthetic evaluation data using GPT-4o. Each example is a single-turn prompt labelled as high-stakes or low-stakes. The out-of-distribution evaluation sets differ structurally: Anthropic HH [1] consists of real multi-turn conversations; ToolACE [9] consists of multi-turn conversations with function/tool calls. Labels for both were assigned by GPT-4o on a 1–10 scale and binarised.

We refer to original-language datasets as "Original" rather than "English" because the training and synthetic test data are multilingual.

Table 1: Evaluation datasets used in this study.

Dataset	Orig size	ID size	Format
Synthetic test	2,000	1,950	Single-turn, multilingual, balanced
Anthropic HH	2,984	2,959	Multi-turn, real-world conversations
ToolACE	734	733	Multi-turn with function/tool calls

Table 2: Architecture details for the two evaluated models.

Property	Llama 3.1 8B	Gemma 3 12B
Layers	32	48
Hidden dim	4,096	3,840
ID support	Tier 2	Multilingual training
Quantisation	8-bit (bitsandbytes)	8-bit (bitsandbytes)

Indonesian translation. We translated all three evaluation sets into Indonesian using Claude Sonnet 4.5 (`claude-sonnet-4-5-20250929`) via the Anthropic API. The system prompt instructed the model to translate from any source language to Indonesian, prioritising native naturalness over literal accuracy, using dictionary-standard Indonesian (KBBI), preserving tone and formality, and keeping proper nouns, technical terms, and code unchanged. For multi-turn dialogues, each turn was translated independently while preserving role structure.

Translation quality was validated by the author (a native Indonesian speaker) across iterative prompt refinements. The Indonesian sets are slightly smaller because some examples were excluded due to translation API refusals.

3.3 Model Details

4 Results

4.1 Layer Sweep

To find where each model encodes the most useful signal for high-stakes detection, we train probes at several layers spanning early to late in the network: five layers for Llama (12, 16, 20, 26, 31 out of 32) and seven for Gemma (8, 16, 24, 31, 32, 41, 47 out of 48). We evaluate each probe on three datasets in both the original language and Indonesian, reporting the mean AUROC across 10 seeds.

Averaging across all three datasets per language, the best overall layers are layer 12 for Llama (38% of network depth, mean AUROC 0.9359) and layer 32 for Gemma (67%, mean 0.9310). On both models, performance on out-of-distribution benchmarks degrades at later layers while synthetic performance remains high across all layers (Figures 1 and 2).

4.2 Cross-Lingual Performance

With the best layers identified, we ask whether the probe still works when inputs are in Indonesian rather than the original language. Table 5 compares AUROC at the best layer for each model on the two out-of-distribution benchmarks.

The gaps are small: 0.9% mean AUROC degradation on Llama, 1.0% on Gemma. The probe reads a representation that is largely language-invariant at these layers.

4.3 Cross-Lingual Agreement

The aggregate AUROC gap is small, but that alone does not tell us whether language affects *which* examples the probe gets wrong. If every example received the same prediction in both languages, we could be confident that language has no effect on the probe's behaviour, even at a per-example level. Conversely, if many examples flip between correct and incorrect across languages, the similar aggregate scores would be coincidental rather than reflecting true invariance. We compare predictions

Table 3: Layer sweep for Llama 3.1 8B (32 layers). Mean AUROC across 10 seeds.

Layer	Orig Synth	Orig Anth	Orig Tool	ID Synth	ID Anth	ID Tool
12	**0.9985**	**0.9358**	0.8734	**0.9970**	**0.9044**	**0.8870**
16	0.9981	0.9077	**0.8773**	0.9964	0.8780	0.8707
20	0.9974	0.8992	0.8540	0.9952	0.8696	0.8607
26	0.9960	0.8649	0.8363	0.9910	0.8236	0.8543
31	0.9944	0.8570	0.8528	0.9852	0.7593	0.8417

Table 4: Layer sweep for Gemma 3 12B (48 layers). Mean AUROC across 10 seeds.

Layer	Orig Synth	Orig Anth	Orig Tool	ID Synth	ID Anth	ID Tool
8	0.9941	0.7872	0.7850	0.9898	0.7836	0.7838
16	0.9988	0.9065	0.8568	0.9977	0.8984	**0.8665**
24	**0.9990**	**0.9342**	0.8531	**0.9982**	**0.9246**	0.8389
31	0.9987	0.9209	0.8542	0.9976	0.9158	0.8452
32	0.9987	0.9238	0.8706	0.9975	0.9176	0.8571
41	0.9979	0.9055	0.8751	0.9953	0.8700	0.8594
47	0.9976	0.8832	**0.8752**	0.9943	0.8811	0.8515

on Anthropic HH example-by-example across the two languages. For Gemma, the remaining analyses use layer 24 rather than layer 32 (the best probe layer). Layer 24 is the second-best probe layer (mean AUROC 0.929 vs. 0.931, a gap of 0.2%), sits at comparable relative depth to Llama's layer 12 (50% vs. 37.5% of total depth), and has a corresponding Gemma Scope 2 SAE with Neuronpedia [11] feature descriptions (Section 5). On Llama, 91.7% of predictions are stable across languages. On Gemma at layer 24, 88.8% are stable, with a translation-induced failure rate of 7.9% compared to Llama's 5.0% (1.6×). The gap between models is smaller than the aggregate AUROC numbers suggest (1.0% vs. 0.9%), but Gemma still shows more per-example sensitivity to language at this layer.

4.4 Error Analysis

Beyond comparing languages, we want to understand whether the probe's mistakes lean in a particular direction: does it more commonly miss genuinely high-stakes interactions (false negatives) or incorrectly flag benign ones (false positives)?

Error direction differs between models. On Llama, errors are overwhelmingly false negatives: 600 FN vs. 51 FP on Anthropic, and 344 FN with zero FP on ToolACE. The probe under-triggers, missing high-stakes content embedded in longer conversations where mean-pooling dilutes the signal.

On Gemma at layer 24, the pattern reverses. Anthropic errors are FP-dominated (253 FP vs. 162 FN), and ToolACE errors are strongly FP-dominated (155 FP vs. 43 FN). The probe over-triggers, classifying benign interactions as high-stakes. This reversal is specific to layer depth: at Gemma's best probe layer (L32), the pattern is FN-dominated like Llama (91 FP vs. 498 FN on Anthropic).

ToolACE on Llama. ToolACE high-stakes scenarios involve tool/function calls where the stakes come from the *consequence* of the action, not from explicit danger language. The Llama probe, trained on conversational data, does not recognise stakes in this format (zero FP, 47% FN rate).

Figure 4 shows that error examples in raw activation space are not geometrically distinct from correct predictions.

5 Decomposing Probe Failures with Sparse Autoencoders

The error analysis shows *what* the probe gets wrong: it misses high-stakes content buried in longer conversations and does not recognise stakes expressed through tool calls. A natural follow-up is

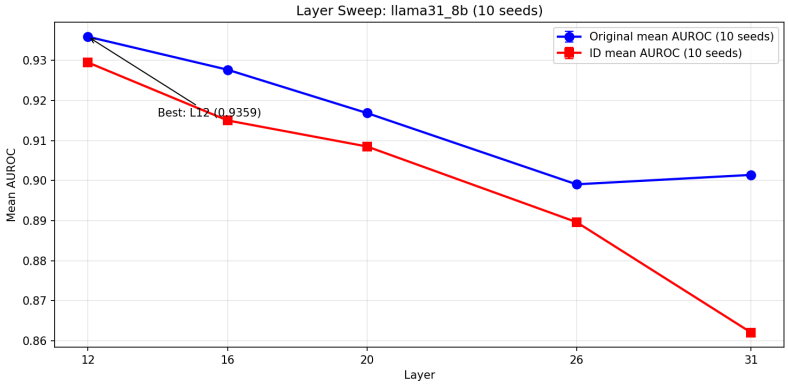


Figure 1: Layer sweep for Llama 3.1 8B. Mean AUROC across Synthetic, Anthropic HH, and ToolACE. Layer 12 is best for both Original (0.9359) and Indonesian (0.9295). The cross-lingual gap (vertical distance between curves) remains roughly constant across layers.

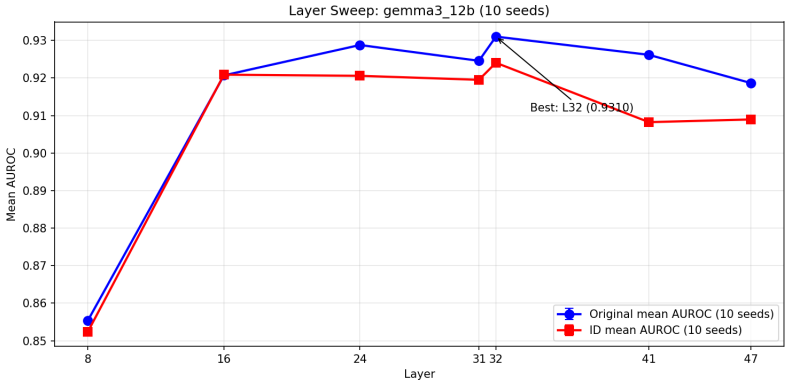


Figure 2: Layer sweep for Gemma 3 12B. Unlike Llama's monotonic decline, Gemma's probe quality rises through the first third of the network before plateauing and declining. Layer 32 is best for both Original (0.9310) and Indonesian (0.9241). The cross-lingual gap widens at deeper layers (41, 47).

whether we can characterise *which* features in the model's representation are associated with these failures.

PCA shows that errors are not geometrically separable from correct predictions in activation space (Figure 4). We therefore use Sparse Autoencoders (SAEs) to decompose activations into individual learned features and check whether any are differentially active in errors.

5.1 Approach

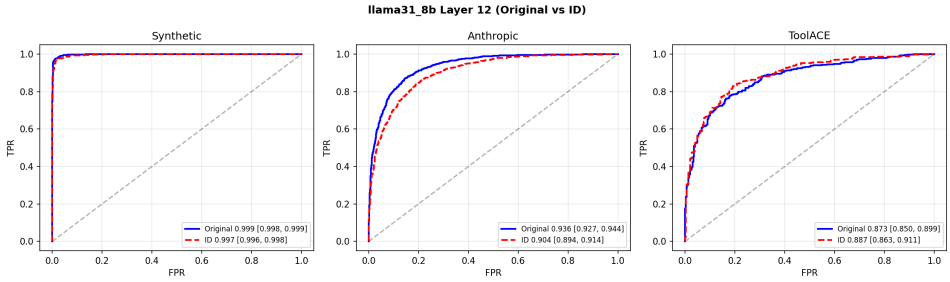
Sparse Autoencoders (SAEs) decompose a model's activation vector into a sparse linear combination of learned feature directions, each ideally corresponding to an interpretable concept [2]. We use pre-trained SAEs from two families:

- **Llama.** Llama Scope [7], 32K features (8× expansion), at layers 12 and 26. These SAEs were trained on the base model; we apply them to the instruct model, which the Llama Scope paper [7] validates in Section 4.5.2: no significant degradation in reconstruction loss or sparsity is observed for layers 0–30.
- **Gemma.** Gemma Scope 2 [5], 16K features, at layers 24 and 41. The best probe layer is 32, but Gemma Scope 2 provides SAEs only at layers {12, 24, 31, 41}; we use layer 24 (below the best layer) and layer 41 (above it) to compare behaviour at different depths.

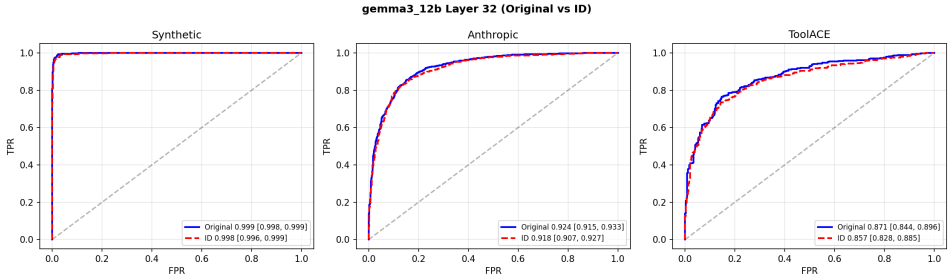
Layer choice for Gemma. For this analysis we use Gemma layers 24 and 41.

Table 5: Cross-lingual AUROC at best layer. Delta = Indonesian − Original.

Model	Test Set	Original	Indonesian	Δ
Llama L12	Anthropic	0.9358	0.9044	−0.0314
	ToolACE	0.8734	0.8870	+0.0136
	Mean (OOD)	**0.9046**	**0.8957**	**−0.0089**
Gemma L32	Anthropic	0.9238	0.9176	−0.0062
	ToolACE	0.8706	0.8571	−0.0135
	Mean (OOD)	**0.8972**	**0.8873**	**−0.0099**



(a) Llama 3.1 8B, layer 12. Blue = Original; red dashed = Indonesian.



(b) Gemma 3 12B, layer 32. Blue = Original; red dashed = Indonesian.

Figure 3: ROC curves on Synthetic, Anthropic HH, and ToolACE. The Synthetic curves are near-identical in both languages (≈0.999). On Anthropic, the Original curve sits visibly above Indonesian for Llama (0.936 vs. 0.904); the gap is smaller for Gemma (0.924 vs. 0.918). On ToolACE, Indonesian marginally exceeds Original on Llama (within bootstrap CIs).

Sequence-level aggregation. SAEs encode individual token activations, but our probe classifies entire sequences. To obtain one feature vector per example, we max-pool each feature's activation across all tokens in the sequence, capturing whether the feature fired *anywhere* in the input.

Differential prevalence. Counting which features appear most often in errors is confounded by features that are universally active (present in both correct and incorrect examples). We compute `error_rate` − `correct_rate` for each feature, isolating those specifically enriched in errors after controlling for base rates.

5.2 Overview

Table 8 summarises the results across both models and all four layers, and Table 9 gives the full L0 breakdown by dataset.

The results split by model. On Llama, the 32K SAEs are sparse enough (L0 as low as 66 at layer 12) for differential analysis to surface features modestly enriched in errors. These correspond to broad topic categories, detailed below. On Gemma, the 16K SAEs produce L0 values above 1,600 at both layers tested, and no feature exceeds 0.6% differential prevalence. The features are too broadly active to distinguish error from correct populations.

Table 6: Per-example prediction agreement on Anthropic HH.

Behaviour	Llama (L12)	Gemma (L24)
Stable predictions	2713 (91.7%)	2628 (88.8%)
Both predicted high-stakes	2163 (73.1%)	2310 (78.1%)
Both predicted low-stakes	550 (18.6%)	318 (10.7%)
Translation-induced disagreements	246 (8.3%)	331 (11.2%)
High-stakes in Orig, low-stakes in ID	149 (5.0%)	235 (7.9%)
Low-stakes in Orig, high-stakes in ID	97 (3.3%)	96 (3.2%)

Table 7: Error counts at analysis layer (Llama L12, Gemma L24), seed 0, threshold 0.5.

Model	Dataset	Errors	Rate	FP	FN
Llama L12	Synthetic	41 / 2000	2.1%	17	24
	Anthropic	651 / 2984	21.8%	51	600
	ToolACE	344 / 734	46.9%	0	344
Gemma L24	Synthetic	33 / 2000	1.7%	13	20
	Anthropic	415 / 2984	13.9%	253	162
	ToolACE	198 / 734	27.0%	155	43

5.3 Llama: Identified Features

Layer 12 (best probe layer, 38% depth). L0 ranges from $\sim$66 (Synthetic) to $\sim$205 (ToolACE) active features per example out of 32K. Table 10 shows the features most enriched in errors on Anthropic HH.

The error-enriched features relate to technology, knowledge, and events, topics where conversations involve consequential actions without explicit danger language. The false-positive features relate to finance and history, topics that sound serious but are often informational. These are domain-level correlates: they tell us which topics cluster with errors, but not why the probe fails on one example within a domain while succeeding on another with similar content.

We also searched for trigger words ("emergency," "urgent," "danger," "kill," "hack") in false-positive texts; none appeared, indicating that false positives at this layer are not driven by individual keywords.

Layer 26 (81% depth). Moving deeper, L0 increases to $\sim$671 on Anthropic, $6\times$ less sparse than layer 12. The probe also makes substantially more errors at this layer (1,024 vs. 651 on Anthropic). Differential features are still identifiable: the top feature (24850, "denial or negative sentiment") reaches $+23.4\%$. But this larger differential reflects the larger error pool rather than sharper signal. The features remain at domain granularity.

5.4 Gemma: No Diagnostic Features at Available Width

Layer 24 (50% depth). L0 $\approx$ 2,600 active features per example out of 16K; 16% of all features fire on every input. Differential analysis finds zero features with more than 0.0% differential. All top features appear at 100% prevalence in both error and correct populations. Neuronpedia labels confirm these are generic: articles ("a, an, the"), number patterns, punctuation, and multilingual token fragments.

Layer 41 (85% depth). L0 decreases to $\approx$ 1,614, sparser than layer 24 but still far above Llama's range. The single candidate feature (11338, labelled as multilingual relational tokens) reaches $+0.6\%$ differential on ToolACE, indistinguishable from noise.

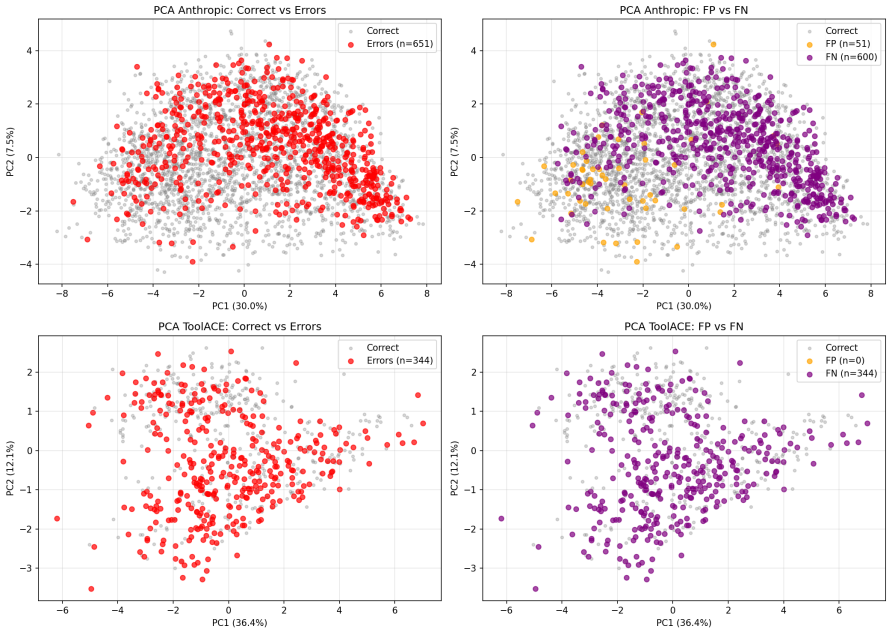


Figure 4: PCA of Llama 3.1 8B activations at layer 12 (Anthropic HH and ToolACE). Error examples (red) scatter throughout the same region as correct examples (grey). The ToolACE FP panel is empty (zero false positives on ToolACE).

Table 8: SAE analysis across layers and models. L0 = mean active features per example after max-pooling.

	Llama (32K SAE)		Gemma (16K SAE)	
Property	L12	L26	L24	L41
L0 (sparsity)	66–205	446–1,140	≈2,600	≈1,614
Max differential	15.3%	23.4%	0.0%	0.6%
Feature granularity	Domain-level	Domain-level	None	None

6 Discussion and Conclusion

Cross-lingual transfer to Indonesian. In our experiments, the linear probe transfers to Indonesian with less than 1% mean AUROC degradation on out-of-distribution benchmarks for both models (Table 5). This is consistent with Wendler *et al.* [12], who show that middle-layer representations in multilingual LLMs converge to a shared semantic space across languages. A linear probe's decision boundary in this shared space would separate "high-stakes" from "low-stakes" regardless of surface language, which matches what we observe.

Per-example agreement differs across models. The aggregate AUROC gaps for Llama (0.9%) and Gemma (1.0%) look similar, but per-example disagreement rates differ. On Llama, 91.7% of Anthropic predictions are stable across languages. On Gemma at layer 24, 88.8% are stable, with 7.9% of examples receiving a correct prediction in the original language but an incorrect one in Indonesian (1.6× Llama's 5.0%; Table 6).

SAE features do not identify specific failure mechanisms. On Llama, the SAE features enriched in probe errors correspond to broad topic categories (technology, finance, knowledge) rather than to properties of the input that would explain why the probe fails (Table 10). They tell us which domains cluster with errors, but not what about a specific input within those domains causes the probe to misclassify it. On Gemma, the SAEs we tested did not produce diagnostic features at either layer (Table 8).

Table 9: SAE sparsity (L0) across layers and models. L0 = mean number of active features per example.

	Llama (32K features)		Gemma (16K features)	
Dataset	L12	L26	L24	L41
Synthetic	66	446	2,597	1,612
Anthropic	112	671	2,602	1,614
ToolACE	205	1,140	2,611	1,615

Table 10: Differentially enriched SAE features on Llama layer 12 (Anthropic HH). Error-enriched rows use all errors (FP+FN combined); FP-enriched rows use false positives only. Differential = error rate − correct rate.

Feature	Error rate	Correct rate	Differential	Neuronpedia label
Error-enriched (all errors)				
6989	59.8%	44.4%	+15.3%	Technical / infrastructure concepts
16646	39.6%	25.4%	+14.3%	Knowledge and information
9766	23.0%	15.1%	+7.9%	Events and happenings
False-positive enriched				
2035	66.7%	45.0%	+21.7%	Financial statistics and reporting
6672	21.6%	8.4%	+13.2%	Historical references

Limitations. We test one language pair (Indonesian) with translated rather than natively-written data. The cross-lingual gap could be larger for languages with non-Latin scripts or less representation in pre-training data. We evaluate only a linear probe on mean-pooled activations; attention probes or other aggregation strategies may show different patterns.

Future work. Extending this work to more models and probing mechanisms beyond mean-pooled linear probes (e.g. attention-weighted pooling) would test whether these findings generalise. Evaluating on natively-written prompts rather than machine-translated ones would also be valuable, since natural language and casual register may differ from translated text in ways that affect probe behaviour.

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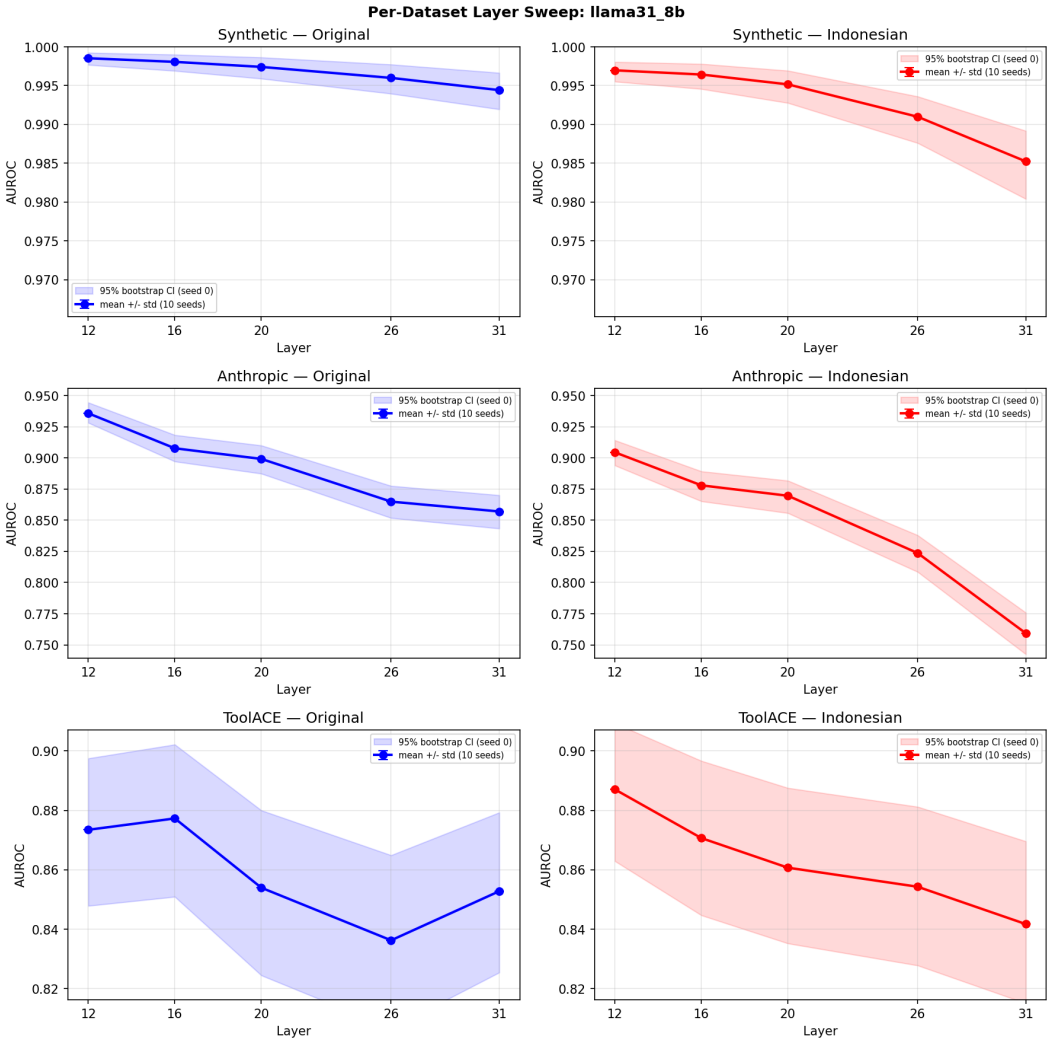

Figure 5: Per-dataset layer sweep for Llama 3.1 8B. Left column: Original. Right column: Indonesian. Shaded region is 95% bootstrap CI (10 seeds). Synthetic performance is high and flat across all layers. Anthropic performance degrades monotonically from layer 12. ToolACE shows high variance due to the small test size (734 examples).

A Per-Dataset Layer Sweep

B Reproducibility Details

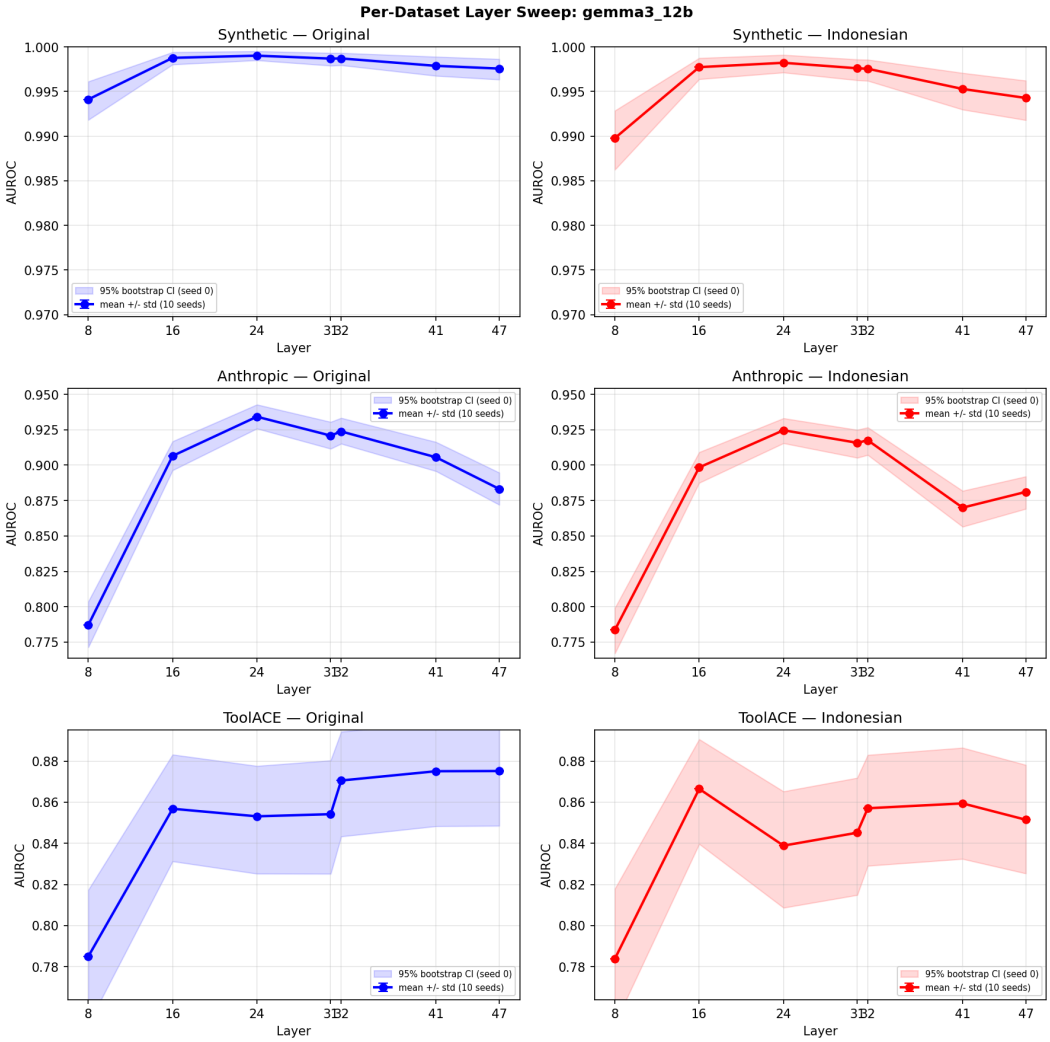

Figure 6: Per-dataset layer sweep for Gemma 3 12B. Left column: Original. Right column: Indonesian. Compared to Llama, Gemma shows a more gradual rise on Anthropic before peaking around layer 32. The Indonesian Anthropic curve shows a larger drop at deeper layers (41, 47) than the Original, reflecting the per-example inconsistency in Section 4.3.

Table 11: Full reproducibility details.

Parameter	Llama 3.1 8B	Gemma 3 12B
Model ID	`meta-llama/Llama-3.1-8B-Instruct`	`google/gemma-3-12b-it`
Quantisation	8-bit (bitsandbytes)	8-bit (bitsandbytes)
Best probe layer	12	32
Hidden dim	4,096	3,840
Activation hook	`model.model.layers[L] .input_layernorm`	`model.language_model .layers[L].input_layernorm`
Pooling	Mean pool over sequence, attention mask	
Max seq. length	8,192 tokens	
BOS handling	Strip leading BOS token	
Probe	`StandardScaler + LogisticRegression(C=1e-3, solver=lbfgs, max_iter=1000, fit_intercept=False)`	
Training set	8,000 examples (`prompts_4x/train.jsonl`)	
Seeds	10 (master seed 20250214)	
SAE (failure analysis)	`fnlp/Llama-Scope` L12R-8x, L26R-8x (32K each)	`google/gemma-scope-2-12b-it` L24, L41 (16K each)
Translation model	`claude-sonnet-4-5-20250929`	
Compute	Lambda Labs A10 GPU (24 GB)	